

Econ. 895 Spatial Techniques in Empirical Economics
George Mason University, Fall 2026
Mondays 7:20 to 10:00 pm
RSCH 201, Fairfax Campus

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Office Hours: By appointment—email me and we will find a time

Course Webpage: Canvas

Course Description

This course teaches the use of spatial and text data in applied economic research. Students will learn to create and analyze spatial data in R, to georeference and digitize archival maps in QGIS, and to construct and validate measures built from text. We will read a selection of articles which use these tools drawn from the Urban, Growth, Trade, Economic History, and Development fields. The ultimate goal of the course is to get students going on empirical research projects that will contribute to their dissertations and eventually be published.

The course teaches these methods alongside the AI tools that now write most first draft code. The fundamentals matter more with large language models rather than less. Verifying what a model did requires the building blocks, and directing a model well requires knowing exactly what to ask for. The central skill this course teaches is verification. You direct the tool, you check its work, and you sign the result. You own every number and every line of code you submit.

The course runs in three arcs. The first five weeks are the fundamentals. You will write vector, raster, and join code yourself, because the building blocks are what make verification possible later, and because the Week 6 lab exam tests whether you can. That exam is closed book and closed model, and it is the only assessment in the course where the tools are off. From Week 6 the emphasis shifts to specifying, verifying, and extending what a model produces. The last three method weeks turn to text as data, because text is where the new tools have changed research practice most.

Week 1 (8/24)	Introduction and the Verification Frame
Week 2 (8/31)	Vector Data, Distance, and IV's
No Class (9/7)	Labor Day

Week 3 (9/14)	Buffers and Regression Discontinuity
Week 4 (9/21)	Rasters and D-in-D
Week 5 (9/28)	Making Data. Georeferencing and Digitizing in QGIS
Week 6 (10/5)	Lab Exam, then LLM's as Data Constructors
No Class (10/12)	Fall Break
Week 7 (10/19)	Network Access and Market Access
Week 8 (10/26)	Climate Data and Spatial Interpolation
Week 9 (11/2)	Cities, Persistence, and Standard Errors in Space
Week 10 (11/9)	Text as Data I. From Corpus to Matrix
Week 11 (11/16)	Text as Data II. Embeddings
Week 12 (11/23)	Text as Data III. LLM's as Measurement
Week 13 (11/30)	Student Research Paper Presentations
Week 14 (12/7)	Student Research Paper Presentations

Note that the weeks above are numbered by class meeting. GMU does not hold Monday classes on Labor Day (September 7) or over Fall Break (October 12), so the course meets fourteen times.

Course Requirements

Eighty percent of success is showing up—Woody Allen

- Research paper worth 50% of your final grade.
- Research paper presentation worth 30% of your final grade. Participation in class discussion, including discussion of the other presentations, folds into this grade.
- Lab exam in Week 6 worth 10% of your final grade.
- Two coding exercises worth 10% of your final grade.

AI Policy

The use of AI tools is permitted and expected on all coursework in this class, including both

coding exercises, the research paper, and the presentation. Every submission must include a short AI use note describing your workflow, meaning what you asked the tools to do and what you checked before signing off. You sign the result and you own every number and every line. Bans on inputs are unenforceable and would only train the honest to lose, so this course polices outputs and accountability instead.

The lab exam in Week 6 is the single exception. It is closed book and closed model, and it tests the material of Weeks 1 to 5 directly.

Nothing stops you from having a model write your code during those first five weeks. The lab exam is what makes writing it yourself worth your while. What you outsource in September you will not be able to do in October, and the exam is exactly where that shows. The building blocks those weeks cover are what make you able to verify a model's work for the rest of your career, and the exam is there to make that argument concrete rather than merely hortatory.

Which Tools, and What They Cost You

From Week 6 the course works with **agentic coding tools**, meaning tools that read your files, write and run code, and iterate on the result, rather than a chat window you paste into. The distinction matters for what this course teaches, because the skill being built is directing and checking a tool that acts. Ferrara's practitioner's guide, assigned in Week 6, sets out the four ways of working with a model and is worth reading before you choose one. I will demonstrate in class using Claude Code.

You are not required to buy anything. George Mason provides PatriotAI free to every student, which gives you access to Claude and ChatGPT models in a data-protected environment, and Microsoft Copilot Chat is included with your Mason Microsoft 365 account. Every graded piece of work in this course can be completed with those, at the cost of more copying and pasting between the model and your editor. Commercial agentic tools run about twenty dollars a month and are considerably more pleasant to work with, but no grade in this course depends on having one, and I will not know which you used unless your AI use note tells me.

If arranging access is a difficulty for you, come and tell me in the first two weeks and we will sort something out. Do not wait until October to raise it.

The Lab Exam

The first part of class in Week 6 is a lab exam completed without LLM access. It certifies the material from the bare hands weeks. You will be asked to carry out and interpret basic spatial operations of the kind covered in Weeks 1 to 5. A student who has worked through the weekly code themselves will find it straightforward.

Coding Exercises

I will distribute two exercises during the semester. Each has two halves. The build half asks you to accomplish a set of spatial tasks in R, with AI assistance permitted and expected, submitted together with your AI use note. The audit half hands you AI generated spatial code that contains planted errors, of the kind that silently corrupt real research, and your grade is for finding them.

Exercise 1 is distributed in Week 6 and due in Week 7. Exercise 2 is distributed in Week 8 and due in Week 9. You will submit executable R code for the build half. I should be able to run your script in a fresh R session without errors, which means you load the necessary packages in the preamble of the code.

Papers

The paper should attempt to test a specific hypothesis using one or more of the empirical tools we discuss in class. Novel data is the strongest differentiator available to you. The judgment calls involved in creating a dataset, and in particular in digitizing archival sources, are precisely the part of research the AI tools cheapen least. The final work should be about 15–20 pages in length, double-spaced in Times Roman 12 pt. font, and must carry the AI use note described above.

If any variable in your paper was constructed by an LLM or another machine classifier, the paper must carry a short **validation appendix**. It needs a gold-standard sample of documents you coded by hand, an agreement statistic between your coding and the machine's (accuracy, F1, or Cohen's kappa), and a note recording the model version and prompts used. Week 12 teaches exactly how to do this, three weeks before the paper is due. A machine-made variable without validation is an unverified number, and you are signing it.

The final draft of the paper must be submitted by midnight on **Monday, December 14th. I will only accept pdf's emailed to me. The subject line of your email should be "Spatial Essay" and the name of the essay pdf should be "your last name Spatial Essay"**. I will take half a letter grade (5%) off your paper grade if you do not hand it in when it is due. I will take an additional half a letter grade off for every 24 hour period you do not hand it in.

Plan also on a short oral spot check near the end of the semester. You will walk me through your own code for about ten minutes and answer one or two questions of the what happens if variety. This is not an interrogation. It is the cheapest honest signal available that the person who signed the code understands it.

I am requiring you to write the paper in L^AT_EX. This is the standard among most researchers in economics today and you might as well learn it now rather than later. You should be able to find many, many, tutorials online for getting started, but here are some I have used:

<http://www.maths.tcd.ie/~dwilkins/LaTeXPrimer/>

Here is another one...

<https://www.tug.org/begin.html>

Presentations

Thirty percent of your course grade is determined by your paper presentation.

I encourage the use of slides for your presentation, but try not to over-do it. If you have questions on what is over-doing it, please refer to this book:

- Tufte, E. (1983). *The visual display of quantitative information*. Graphics Press

I will grade your presentations on “content” (how well you have framed and answered your research question) and “style” (how well you present the material).

Since you are writing the paper in L^AT_EX, you should probably also make your presentation slides using the L^AT_EX presentation environment known as Beamer. Here are some sample slides and tips on how to make a good presentation:

<https://paulgp.github.io/2018/04/30/beamer-tips.html>

The Toolkit for Geospatial Research and Recommended Texts

We will almost exclusively focus on using R as a GIS for reproducible spatial computations. This allows us to use a general purpose programming language to work with spatial data directly, in a powerful statistics environment. We will use QGIS only when we want to georeference and digitize our own maps. Digitizing happens in QGIS and analysis happens in R. For vector data we will be using the simple features paradigm from the *sf* package. For raster data we will be using the *terra* package. In addition to lecture notes and code examples, selected chapters from these books will guide you:

- (the *sf* paradigm) Lovelace, R., Nowosad, J., and J. Muenchow. Geocomputation with R. CRC Press. The second edition is [available online](#).
- (for R in general) Wickham, H., Çetinkaya-Rundel, M., and G. Grolemund. 2023. R for Data Science. 2nd ed. O'Reilly. [Available online](#).
- (for the text weeks) Silge, J. and Robinson, D. (2017). *Text mining with R: A tidy approach*. O'Reilly Media. Online for free [here](#).
- (for the text weeks) Grimmer, J., Roberts, M. E., and Stewart, B. M. (2022). *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press. The framework text behind Weeks 10 to 12. Not distributed with the course readings, so order a copy or use the library. Chapters 5 and 11 support Week 10, chapters 8 and 14 support Week 11, and chapters 15, 18, and 20 support Week 12.
- (useful intro to basic R and econometrics using R) Nick Huntington-Klein's videos at: <http://nickchk.com/videos.html#rstats>

Most of the readings we will discuss can be downloaded from either JSTOR, the NBER Working Papers archive, EconLit, or the course webpage. You are required to acquire these papers and read them before the week they are listed on the syllabus.

Most of the articles require knowledge of econometric techniques. I will attempt to explain the methods used when they come up, however, you may wish to have a more detailed treatment of the methods at your disposal. An excellent source for the basics (plus some more advanced material) is:

- Joshua Angrist and Jorn-Steffen Pischke, *Mastering Metrics: The Path from Cause to Effect*. (Princeton University Press, 2014).

Two more recent books are:

- Cunningham, S. (2020). *Causal Inference: The Mixtape*.
- Huntington-Klein, N. (2021). *The Effect: An Introduction to Research Design and Causality*.

I have also been enjoying this book as a guide to integrating theory with empirical design:

- Ashworth, S., Berry, C.R. and de Mesquita, E.B., (2021). *Theory and Credibility: Integrating Theoretical and Empirical Social Science*. Princeton University Press.

Some good resources on how to write an academic paper are:

- McCloskey, D. (2000). *Economical Writing*. Waveland Press, Incorporated
- Cochrane Writing Tips for PhD Students (in the course readings folder)
- Harvard Sophomore Writing Economics (in the course readings folder)

I also highly recommend this book:

- William Thomson, *A Guide for the Young Economist*. (MIT Press: 2011).

Copyright Notice

Please do not share the material from this course widely. Some of the content in this course derives from material created by Richard Bluhm (Leibniz University) and used with his permission. He, in turn, has the following copyright notice attached to his course materials:

Many sources are not yet properly attributed on the lecture slides, so please to not share these materials widely. I owe a great debt to those who have developed GIS classes for economics audiences before: Masayuki Kudamatsu's IIES course for ArcGIS has been an inspiration to many classes including this one, but I have also used materials from a class Stelios Michalopoulos gave in UWarwick many years ago and a class previously taught Paul Raschky at U. St. Gallen.

The text-as-data material in Weeks 10 to 12 draws on Arthur Spirling's *Text as Data* course at the NYU Center for Data Science, whose lecture materials he has made publicly available at <https://github.com/ArthurSpirling/text-as-data-class-spring2021>. I am grateful for it. Where I have adapted his treatment the errors are mine and the same request applies: please do not circulate these slides widely.

Some Important Dates

First Day of Classes: **8/24**

Labor Day, no class: **9/7**

Fall Break, no class: **10/12**

Thanksgiving Recess: **11/25–11/27**

Last Day of Classes: **12/7**

Final Paper Due: **12/14**

PLEASE NOTE: COURSE POLICIES

Statements 1 through 5 below are the University's Common Policy Addendum, which must be made available to students in all Mason courses (Catalog Policy AP.2.5).

1. Academic Standards

Academic Standards exist to promote authentic scholarship, support the institution's goal of maintaining high standards of academic excellence, and encourage continued ethical behavior of faculty and students to cultivate an educational community which values integrity and produces graduates who carry this commitment forward into professional practice. Our standards are upheld in three principles:

- *Honesty*: providing accurate information in all academic endeavors, including communications, assignments, and examinations.
- *Acknowledgement*: giving proper credit for all contributions to one's work, through accurate citations and references for any ideas, words, or materials created by others.
- *Uniqueness of Work*: ensuring that all submitted work is the result of one's own effort and is original, including free from self-plagiarism.

Violations of these standards — including but not limited to plagiarism, fabrication, and cheating — are taken seriously and will be addressed in accordance with university policies. See <https://academicstandards.gmu.edu/>.

Student responsibility: students are responsible for understanding how these general expectations apply to each course, assignment, or exam; ask me for clarification on any aspect that is not clear.

The AI policy stated earlier in this syllabus is the honor policy for this course as it applies to AI assistance. Work submitted under your name with the required AI use note is your work, and you are accountable for all of it. Work submitted without the required disclosure, or AI assisted work submitted for the lab exam, is an honor code matter.

2. Class Registration

Students are responsible for verifying the accuracy of their own schedules. Students need to check PatriotWeb regularly to verify that they are registered for the classes that they think they are. This is particularly important since students are no longer dropped for nonpayment.

Faculty may not allow a student who is not registered to continue to attend class and may not grade the work of students who do not appear on the official class roster.

Deadlines each semester are published in the Schedule of Classes, available from the Registrar's website: <https://registrar.gmu.edu>.

After the last day to drop a class, withdrawing from this class requires the approval of the dean and is only allowed for nonacademic reasons.

3. Accommodations for Students with Disabilities

Disability Services at George Mason University is committed to upholding the letter and spirit of the laws that ensure equal treatment of people with disabilities. Under the administration of

University Life, Disability Services implements and coordinates reasonable accommodations and disability-related services that afford equal access to university programs and activities. Students can begin the registration process with Disability Services at any time during their enrollment. If you are seeking accommodations, please visit <https://ds.gmu.edu/> for detailed information about the registration process. Disability Services is located in Student Union Building I (SUB I), Suite 2500. Email: ods@gmu.edu. Phone: (703) 993-2474.

Student responsibility: students are responsible for registering with Disability Services and communicating about their approved accommodations with me in advance of any relevant class meeting, assignment, or exam.

4. FERPA and Use of Mason Email Addresses for Course Communication

The Family Educational Rights and Privacy Act (FERPA) governs the disclosure of education records for eligible students and is an essential aspect of any course. Students must use their Mason email account to receive important University information, including communications related to this class. Instructors will not respond to messages sent from, or send messages regarding course content to, a non-Mason email address.

Student responsibility: students are responsible for checking their Mason email regularly for course-related information, or ensuring that Mason email messages are forwarded to an account they do check.

5. Title IX Resources and Required Reporting

As a part of George Mason University's commitment to providing a safe and non-discriminatory learning, living, and working environment for all members of the University community, the University does not discriminate on the basis of sex or gender in any of its education or employment programs and activities. Accordingly, all non-confidential employees, including your faculty member, have a legal requirement to report to the Title IX Coordinator all relevant details obtained directly or indirectly about any incident of Prohibited Conduct (such as sexual harassment, sexual assault, gender-based stalking, dating/domestic violence). Upon notifying the Title IX Coordinator of possible Prohibited Conduct, the Title IX Coordinator will assess the report and determine if outreach is required. If outreach is required, the individual the report is about (the "Complainant") will receive a communication, likely in the form of an email, offering that person the option to meet with a representative of the Title IX office.

For more information about non-confidential employees, resources, and Prohibited Conduct, please see University Policy 1202: *Sexual and Gender-Based Misconduct and Other Forms of Interpersonal Violence*. Questions regarding Title IX can be directed to the Title IX Coordinator via email to TitleIX@gmu.edu, by phone at 703-993-8730, or in person on the Fairfax campus in Aquia 373.

Student opportunity: if you prefer to speak to someone confidentially, please contact one of Mason's confidential employees in Student Support and Advocacy (SSAC), Counseling and Psychological Services (CAPS), Student Health Services (SHS), or the Office of the University Ombudsperson.

6. Campus Closure

If the campus closes or class is canceled due to weather or other concern, students should check the class webpage and monitor their email for updates on how to continue learning and information

about any changes to events or assignments.

Course Outline

Week 1 (8/24): Introduction and the Verification Frame

- Introduction. Why this course still teaches code in 2026, and what verification means in practice. R spatial setup.
- (1) David Weil's Peptalk (in Readings Folder)
- (2) Donaldson, D. and Storeygard, A. (2016). The view from above: Applications of satellite data in economics. *Journal of Economic Perspectives*, 30(4):171–98
- (3) Michalopoulos, S. and Papaioannou, E. (2018). Spatial patterns of development: A meso approach. *Annual Review of Economics*, 10:383–410
- Work through the videos on “Introduction to R for Economists” and “Basic R Videos” at <http://nickchk.com/videos.html#rstats>

Week 2 (8/31): Vector Data, Distance, and IV's

- The building blocks of vector data in R. Points, lines, and polygons as simple features, then attribute and spatial operations, then measuring distance in a projected world. We finish by building Dittmar's instrument, distance from Mainz, from raw coordinates.
- (4) Dittmar, J. E. (2011). Information technology and economic change: the impact of the printing press. *The Quarterly Journal of Economics*, 126(3):1133–1172
- Lovelace et al., chapters 2, 3, and 4, and chapter 7 on reprojection. Read the vector sections. Sections 2.4 and 2.5 on coordinate reference systems and units, and 4.2.3 and 4.2.5 on distance relations and spatial joins, are the ones this week's class runs on.

Week 3 (9/14): Buffers and Regression Discontinuity

- Constructive geometry gives us buffers, centroids, and clipping. A boundary becomes a research design when you can measure distance to it. We build Dell's mita discontinuity to see how.
- (5) Dell, M. (2010). The persistent effects of Peru's mining mita. *Econometrica*, 78(6):1863–1903
- Lovelace et al., chapter 5, section 5.2 on geometric operations for vector data. Section 5.2.3 covers buffers and 5.2.5 covers clipping.

Week 4 (9/21): Rasters and D-in-D

- The other half of spatial data. Grids rather than shapes, map algebra, and the extract step that turns a raster into a column in your regression. Nunn and Qian's potato suitability measure is exactly that operation.

- (6) Nunn, N. and Qian, N. (2011). The potato's contribution to population and urbanization: Evidence from a historical experiment. *The Quarterly Journal of Economics*, 126(2):593–650
- Lovelace et al., the raster sections of chapters 2 to 5, meaning 2.3, 3.3, 4.3, and 5.3, and then chapter 6 on raster-vector interactions. Section 4.3.2 on map algebra and 5.3.4 on resampling are the ones that bite.

Week 5 (9/28): Making Data. Georeferencing and Digitizing in QGIS

- Georeferencing printed archival maps, choosing control points, and tracing features. Digitizing happens in QGIS and analysis happens in R. This is the week to start thinking seriously about what novel data your paper could be built on.
- QGIS Documentation, the Georeferencer chapter, [available online](#).

Week 6 (10/5): Lab Exam, then LLM's as Data Constructors

- Lab exam in the first part of class. No LLM access.
- Then the tools come on. Geocoding messy historical place names and extracting entities from archival text into coordinates. A model's output is a dataset that needs validation like any other dataset.
- (7) Ferrara, A. (2026). A practitioner's guide to using large language models and generative AI in economic history. NBER Working Paper 35374, National Bureau of Economic Research
- Recommended: Korinek, A. (2023). Generative AI for economic research: Use cases and implications for economists. *Journal of Economic Literature*, 61(4):1281–1317. The same genre as Ferrara but for economic research generally. Read the two together and you have the current playbook.
- Coding Exercise 1 Distributed

Week 7 (10/19): Network Access and Market Access

- Distance stops being a straight line. Least cost paths across a network, and market access as a summary of the whole network at once. In class we will extend working least-cost path code with AI assistance, and check what it did.
- (8) Donaldson, D. and Hornbeck, R. (2016). Railroads and American economic growth: A “market access” approach. *The Quarterly Journal of Economics*, 131(2):799–858
- (9) Johnson, N. D. and Koyama, M. (2017). Jewish communities and city growth in preindustrial Europe. *Journal of Development Economics*, 127:339–354
- Coding Exercise 1 Collected

Week 8 (10/26): Climate Data and Spatial Interpolation

- Measuring something everywhere when you observed it only in a few places. Interpolation turns scattered station readings into a surface, and every such surface is a model whose errors you inherit.
- (10) Anderson, R. W., Johnson, N. D., and Koyama, M. (2017). Jewish persecutions and weather shocks: 1100–1800. *The Economic Journal*, 127(602):924–958
- These two tutorials on spatial interpolation are useful:
 - https://mgimond.github.io/Spatial/16_interpolation.html
 - <https://rspatial.org/analysis/4-interpolation.html>
- Coding Exercise 2 Distributed

Week 9 (11/2): Cities, Persistence, and Standard Errors in Space

- Persistence claims are sharpest in the cities literature, which is also where spatial correlation does the most damage to a standard error. Conley and Kelly reanalyze the portage result directly. In class we will replicate a figure from one of these papers with AI assistance and check what the model did.
- (11) Davis, D. R. and Weinstein, D. E. (2002). Bones, Bombs, and Break Points: The Geography of Economic Activity. *American Economic Review*, 92(5):1269–1289
- (12) Bleakley, H. and Lin, J. (2012). Portage and path dependence. *The Quarterly Journal of Economics*, 127(2):587–644
- (13) Conley, T. G. and Kelly, M. (2025). The standard errors of persistence. *Journal of International Economics*, 153:104027
- (14) Colella, F., Lalive, R., Sakalli, S. O., and Thoenig, M. (2019). Inference with arbitrary clustering. IZA Discussion Paper 12584, IZA Institute of Labor Economics
- Coding Exercise 2 Collected

Week 10 (11/9): Text as Data I. From Corpus to Matrix

- Turning a pile of documents into a matrix you can regress on, and seeing how much of the answer your own preprocessing decided. In class we will run one corpus through two defensible pipelines and watch the results change. Folklore is the bridge paper. Text from oral traditions becomes measures of culture, joined to ethnic homelands spatially.
- (15) Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–74

- (16) Denny, M. J. and Spirling, A. (2018). Text preprocessing for unsupervised learning: Why it matters, when it misleads, and what to do about it. *Political Analysis*, 26(2):168–189
- (17) Michalopoulos, S. and Xue, M. M. (2021). Folklore. *The Quarterly Journal of Economics*, 136(4):1993–2046
- Recommended: Grimmer et al. (2022), chapters 5 and 11. See Recommended Texts.

Week 11 (11/16): Text as Data II. Embeddings

- Meaning as geometry and what a vector space buys and hides. The paper applies the market-access tools of Week 7 to the print world of Week 2. Because the researcher’s choices now sit inside the model, verification becomes probing. In class we will change the seed and context window behind its title embeddings and check which nearest neighbors survive.
- (18) Johnson, N. D. and Taylor, A. (2026). Mapping the market for ideas in Europe, 1450–1650: A title-embeddings approach. Working paper, George Mason University
- Recommended: Grimmer et al. (2022), chapters 8 and 14. See Recommended Texts.

Week 12 (11/23): Text as Data III. LLM’s as Measurement

- Two conditions have to hold before a machine-made variable can be trusted. The first is no training leakage. The second is validation against a gold standard you coded yourself. In class we will hand-code a sample of book titles from my work in progress on plague, print, and persecution, and compute agreement against the classifier live. This is the week that teaches the validation appendix your paper requires.
- (19) Ludwig, J., Mullainathan, S., and Rambachan, A. (2026). Large language models: An applied econometric framework. *Annual Review of Economics*, 18:283–316
- Recommended: Dell, M. (2025). Deep learning for economists. *Journal of Economic Literature*, 63(1):5–58. A reference survey of the whole deep learning toolkit. Keep it on the shelf.
- Recommended: Grimmer et al. (2022), chapters 15, 18, and 20. See Recommended Texts.

Week 13 (11/30): Student Research Paper Presentations

- You present your paper and defend its choices. Come prepared to say what you would have to believe for your identification to hold, and how you checked the numbers you are showing us.

Week 14 (12/7): Student Research Paper Presentations

- The second half of the presentations, on the same terms. The final paper is due the following Monday, December 14th.

References

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- Ferrara, A. (2026). A practitioner’s guide to using large language models and generative AI in economic history. NBER Working Paper 35374, National Bureau of Economic Research.
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